

AeroMind: Airport Operations Intelligence Platform

Abstract

The aviation industry creates an amount of operational data every day. To improve airport performance, reduce delays and make airline operations better we need to analyze this data. Traditional reporting methods usually focus on what happened in the past and do not help much with making decisions about what to do.

The goal of this project was to create “**AeroMind**” a platform that turns flight data into useful information for businesses. We followed an analytics process starting with understanding the business and going through data cleaning designing a data warehouse analyzing data with SQL creating new features exploring data using machine learning and building a dashboard.

We used a dataset that had information about flights in the year 2024 including schedules, delays, cancellations and reasons for delays. We used Python to prepare the data and to build a data warehouse and MySQL to run queries and ML to find patterns in operations. Finally we created a dashboard to visualize important metrics and support decision-making.

This project shows that I have skills in data analytics, business intelligence, SQL, machine learning, building dashboards and using AI to develop software.

1. Introduction

Modern airports are busy places where thousands of flights are coordinated every day. Delays, cancellations and congestion can affect how well airlines and airports work and how happy passengers are.

Airports collect a lot of data but they need to analyze it in a structured way to get useful insights. BI platforms help airport managers monitor performance, find patterns in delays, understand congestion and evaluate airline performance using data.

The AeroMind project was created to show how a real-world airport operations analytics solution can work by combining data engineering, SQL analytics, machine learning and dashboard visualization.

2. Project Objectives

The main goals of this project were:

- Analyze airport performance using flight data.
- Find the reasons for flight delays and disruptions.
- Build a data warehouse using a Star Schema.
- Do business analysis using SQL.
- Create features to improve analysis.
- Develop ML models for operations analysis.
- Build a dashboard to support airport management decisions.
- Show an end-to-end analytics process that can be used in companies.

3. Dataset Overview

The project uses the [Flight Delay Dataset 2024](#) which has information about flights, including airline details, airport information, departure and arrival times, delays, cancellations and reasons for delays.

The dataset includes:

- Flight schedules
- Airline information
- Airport information
- Departure and arrival delays
- Cancellation information
- Diversion information
- Reasons for delays

We created a data dictionary to document each field's type, description and business meaning.

4. Data Preparation

We prepared the data using Python. The main activities were:

- Handling missing values
- Correcting data types
- Removing duplicate records
- Creating business- columns
- Standardizing values
- Creating route information
- Creating scheduled hour features
- Creating delay status indicators

We did an EDA to identify missing values, understand delay distributions, analyze airline coverage, evaluate airport coverage and detect operational trends.

5. Data Warehouse Design

We designed a data warehouse based on a Star Schema to organize the operational data efficiently.

The warehouse has one central fact table and five dimension tables:

- Dim_Airline
- Dim_Airport
- Dim_Date
- Dim_Time
- Dim_Delay_Type
- Fact_Flight_Operations

The Star Schema separates information from measurable operational data making queries faster and reporting easier. We established foreign key relationships to support analytical queries.

6. SQL Analytics

We used SQL to do business-oriented analysis on the warehouse. The analytical queries were divided into categories:

- **KPIs:** Total flights, average departure delay average arrival delay, cancellation rate, diversion rate
- **Airport Analytics:** Airports with highest delays, best performing airports, airport operational rankings
- **Airline Analytics:** Airline delay rankings, cancellation rankings, diversion analysis
- **Congestion Analytics:** Peak operational hours, high traffic periods, congestion trends
- **Root Cause Analytics:** evaluating the contribution of different delay categories

These SQL analyses formed the foundation for executive reporting and dashboard development.

7. Feature Engineering & Machine Learning

We created features to improve analytical capability beyond the original dataset. These included:

- Airport Flight Volume
- Route Information
- Scheduled Hour
- Delayed Flight Indicator
- Total Delay Causes
- Extreme Delay Flag

We applied ML techniques to the engineered dataset to identify patterns & evaluate predictive performance. The workflow included data preprocessing, feature selection, model training, model evaluation and prediction visualization.

8. Dashboard Development

We developed a dashboard called “**AeroMind Command Center**” to present business insights through interactive visualizations.

The dashboard includes:

- Operational KPI Cards
- Flight Performance Overview
- Airport Performance Analysis
- Airline Performance Analysis
- Delay Distribution
- Congestion Monitoring
- Root Cause Analysis
- Operational Summary

The dashboard was designed for an audience enabling users to quickly identify operational issues and monitor airport performance.

9. Project Outcomes

The project shows how historical flight data can be transformed into operational intelligence.

The project successfully built a data warehouse performing SQL analytics, engineered meaningful operational features, conducted exploratory and statistical analysis, developed machine learning models and created an executive-level airport operations dashboard.

10. Conclusion

AeroMind integrates data engineering, business intelligence statistical analysis, machine learning and dashboard visualization into an airport operations analytics platform.

The project emphasizes the analytics lifecycle, from business understanding, to executive decision support. The structured data warehouse, SQL analytics, feature engineering process and interactive dashboard demonstrate skills required for modern Data Analyst and Business Analyst roles.

This project has strengthened my understanding of data preparation, relational database design, SQL analytics, EDA, ML and business storytelling. It highlights the use of AI-assisted development tools to improve productivity while maintaining ownership of the analytical process.

Overall AeroMind serves as a portfolio project that reflects industry-oriented thinking and demonstrates the ability to solve real-world business problems using data.